

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

</tool_call> reasoning process here

The de Broglie wavelength λ of a particle is given by :

$$\lambda = h / p$$

where :

- h is Planck's constant = 6.626×10^{-34} J $\cdot$ s (or 6.626×10^{-27} erg $\cdot$ s),
- p is the momentum of the particle .

For a non-relativistic electron , kinetic energy K is related to momentum by :

$$K = p^2 / (2m)$$

$$\text{So , } p = \sqrt{2mK}$$

Thus , the de Broglie wavelength becomes :

$$\lambda = h / \sqrt{2mK}$$

$$\text{We are given } K = 1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$\text{Mass of electron } m = 9.109 \times 10^{-31} \text{ kg}$$

Plug in values :

$$\lambda = (6.626 \times 10^{-34}) / \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}}$$

First compute denominator :

$$2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19} = 2 \times 9.109 \times 1.602 \times 10^{-50} \\ = 28.82 \times 10^{-50} = 2.882 \times 10^{-49}$$

$$\sqrt{2.882 \times 10^{-49}} = \sqrt{2.882 \times 10^{-24.5}} \approx 1.698 \times 10^{-24.5}$$

$\sqrt{1.698 \times 10^{-24.5}}$ better to compute numerically :

$$\sqrt{2.882 \times 10^{-49}} = \sqrt{2.882} \times 10^{(-24.5)} \approx 1.698 \times 10^{(-24.5)}$$

$$\text{But } 10^{(-24.5)} = 10^{(-24)} \times 10^{(-0.5)} \approx 10^{(-24)} \times 0.03162$$

$$\text{So } \sqrt{2.882 \times 10^{-49}} \approx 1.698 \times 0.03162 \times 10^{-24} \approx 5.38 \times 10^{-26}$$

Wait $\sqrt{2.882}$ better step-by-step :

$$\sqrt{2.882 \times 10^{-49}} = \sqrt{2.882} \times \sqrt{10^{-49}} = 1.698 \times 10^{(-24.5)}$$

$$10^{(-24.5)} = 10^{(-24)} \times 10^{(-0.5)} = 10^{(-24)} \times 0.03162$$

$$\text{So : } 1.698 \times 0.03162 \times 10^{-24} \approx 0.0538 \times 10^{-24} = 5.38 \times 10^{-26}$$

Wait $\sqrt{2.882}$ mistake in exponent .

$$\text{Wait : } \sqrt{10^{-49}} = 10^{(-49/2)} = 10^{(-24.5)} = 10^{(-24)} \times 10^{(-0.5)} = 10^{(-24)} \times 3.162 \times 10^{-2} \approx 3.162 \times 10^{-26}$$

$$\text{Then } \sqrt{2.882} \approx 1.698$$

$$\text{So } \sqrt{2.882 \times 10^{-49}} = 1.698 \times$$